5.1 Introduction

A dc machine is a device that can convert electrical energy into mechanical energy or mechanical energy into electrical energy. Thus dc machine is an energy conversion device. When the machine converts mechanical energy into dc electrical energy; it is called dc generator. When the machine converts dc electrical energy into mechanical energy; it is called dc motor. In principle, a dc machine can be made to work as a dc motor and as a dc generator depending upon the input energy supplied. However, the fundamental principles involved in the operation of motor and generator are different. In the following sections principles of dc machines are explained.

5.2 Principles of DC Machines

A dc machine essentially consists of a system of conductors and system of electromagnets (magnets). The system of electromagnet is called the field system and it produces the required working flux. The field system is stationary. The system of conductors is called armature and it is mounted on a movable shaft.

The armature is electrically connected to the two terminals of an external circuit through some mechanism. In dc generator the mechanical energy is supplied to the machine is the form of rotational energy of the shaft and the electrical energy is taken as shown in fig. 5.1 (b). In dc motor, the electrical energy is supplied to the machine through the two terminals and mechanical output is taken from the shaft in the form of rotational energy of the shaft as shown in fig. 5.1 (a).

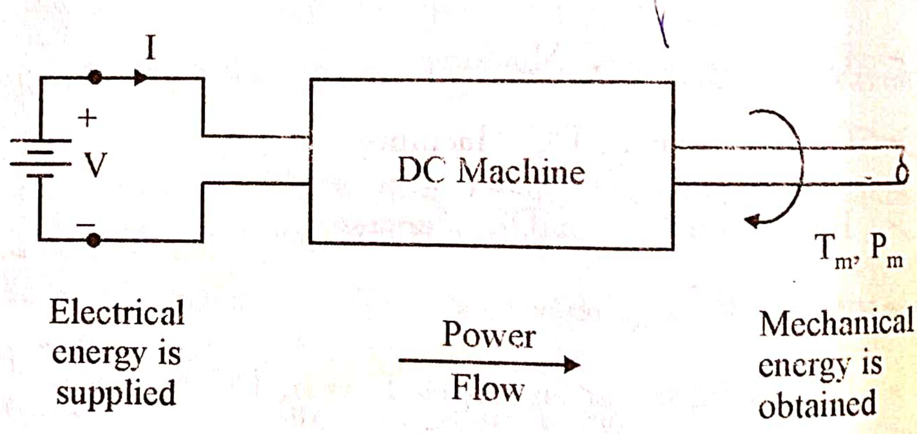

Fig. 5.1 (a) DC machine as Motor

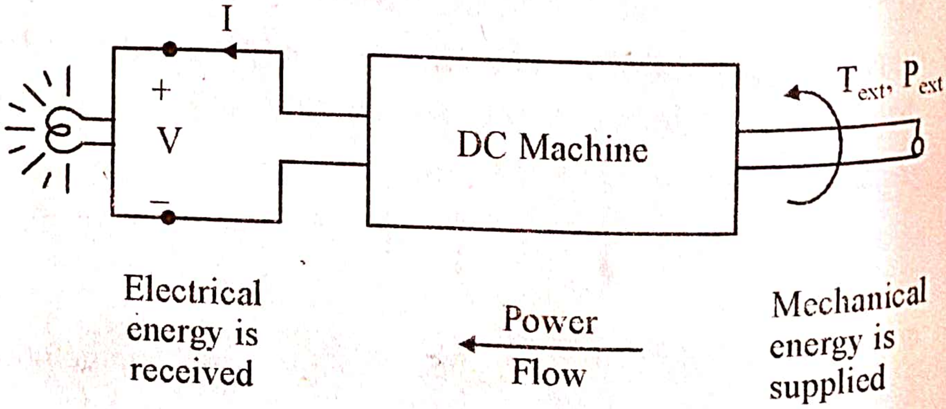

Fig. 5.1 (b) DC machine as Generator

5.2.1 Principle of DC Generator

The principle of dc generator is based on Faraday's law of electromagnetic induction. This law states that whenever there is change in flux linkage of a coil, an emf is induced in the coil.

The magnitude of this induced emf is given by

$$e = \frac{d\phi}{dt}$$

where, ϕ is the flux linkage of the coil. In dc generator, flux is stationary and the coil rotates, therefore induced emf is dynamic induced emf (refer section 4.3).

The working principle of dc generator can be explained with the help of a single loop generator as shown in fig. 5.2 When the coil is rotated by some prime mover (mechanical energy), there is a change in flux linkage of the coil. If ϕ is the average flux produced by each magnetic pole, the average induced emf can be obtained as

$$E_{av} = \frac{d\phi}{dt} = \text{flux cut per sec by the coil}$$

$$= (\text{flux cut in one rotation}) \times \text{rotation per sec}$$

$$= (\text{flux per pole} \times \text{no. of poles}) \times \text{rotation per sec}$$

$$\boxed{E_{av} = \phi p n \text{ Volts}} \qquad \ldots(5.1)$$

Where, n is the speed of coil in rps and p is the number of poles in the generator.

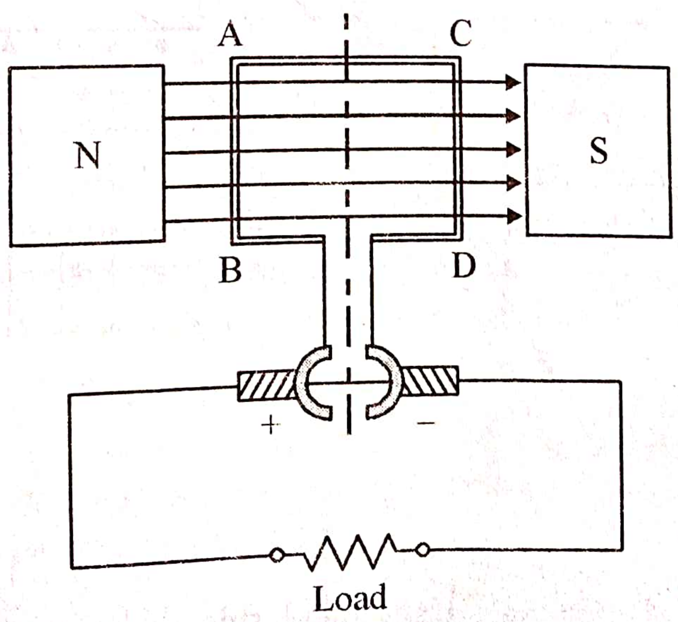

Fig. 5.2 Single loop dc generator

Alternatively; the average induced emf per conductor can be found as

$$E_{av/conductor} = B\,\ell\,v \text{ Volts} \qquad \text{....(5.2)}$$

Where, B average flux density (wb/m^2)

ℓ length of the conductor (m)

v velocity of conductor (m/sec)

Since the two conductors of the coils are under the influence of opposite poles, the net induced emf in the coil is the sum of induced emfs of the two conductor

$$\boxed{E_{av} = 2B\ell V} \text{ Volts} \qquad \text{.....(5.3)}$$

It is important to note that expression (5.1) and (5.3) are identical expressions. One expression can be derived from the other expression.

The direction of dynamic induced emf in a conductor is given by Fleming's Right Hand Rule. According to this rule *"if the thumb, forefinger and middle finger of right hand are kept mutually perpendicular to each other and if fore finger points to the direction of flux, thumb points to the direction of motion, then the middle finger will point to the direction of induced emf."*

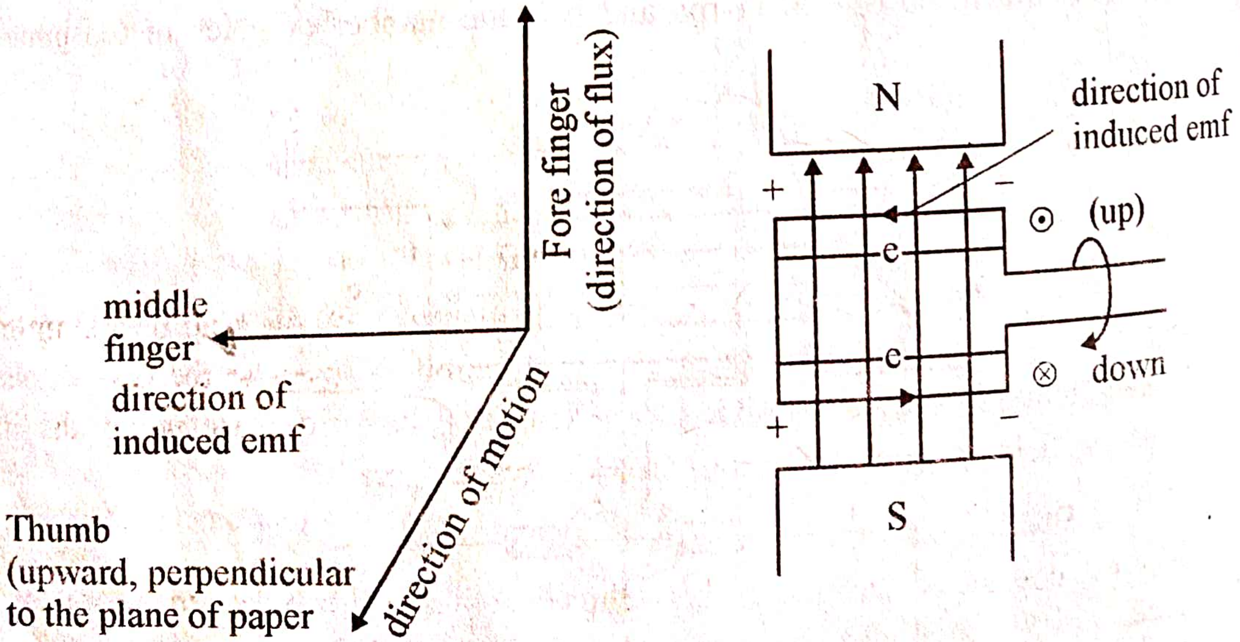

Fig. 5.3 Fleming's right hand rule & its application

5.2.2 Principle of DC Motor

The principle of dc motor is based on the law of electromagnetic interaction. According to this law, whenever a current carrying conductor is placed in a magnetic field, the conductor experiences an electromagnetic force. The magnitude of this force is given by

$$\boxed{F = BI\ell} \text{ Nt} \qquad .....(5.4)$$

Where I is the current flowing in the conductor (A)

ℓ is length of the conductor (m)

B flux density of the magnetic field (wb/m^2)

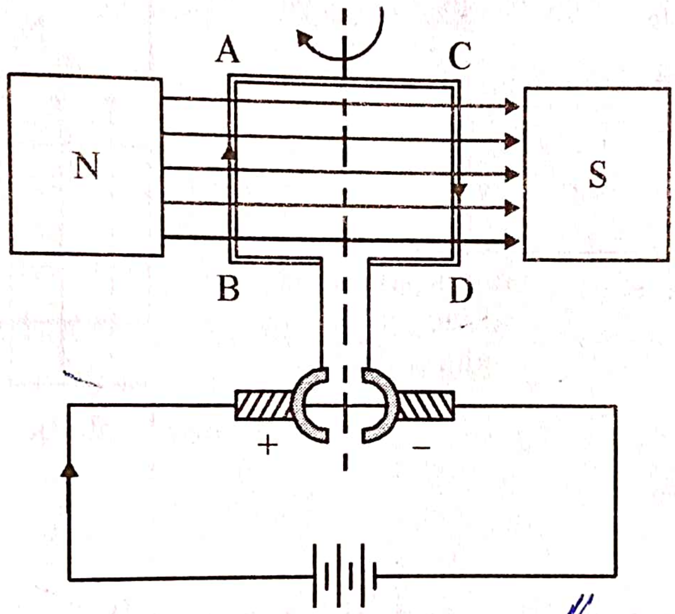

Fig. 5.4 Single loop dc motor

The working principle of dc motor can be explained with the help of a simple single coil motor as shown in fig. 5.4. When electrical supply is given to the coil. current I_a starts flowing in both the sides of the coil. Since the coil sides are under the influence of magnetic field B, each coil sides experiences a force

$$F = BI_a\ell$$

The two coil sides are under the influence of opposite poles, therefore the developed forces will be equal and opposite and at a distance d. Thus, they developed a torque in the coil which can be expressed as

$$T = BI_a \ell\, d$$

$$\boxed{T = BI_aA} \qquad .....(5.5)$$

Where, d is the distance between two coil sides

$A = \ell\, d$ is the area of the coil

The torque developed in the coil, rotates the coil and hence we get mechanical energy in the form of rotation of the coil

The direction of induced force in a conductor is given by Fleming's left Hand Rule. This rule states that if the thumb, *"fore finger and middle finger of left hand are kept mutually perpendicular to each other and if fore finger points to the direction of flux, middle finger points to the direction of current, the thumb will point to the direction of force."* Fig. 5.5 explains the left handle rule.

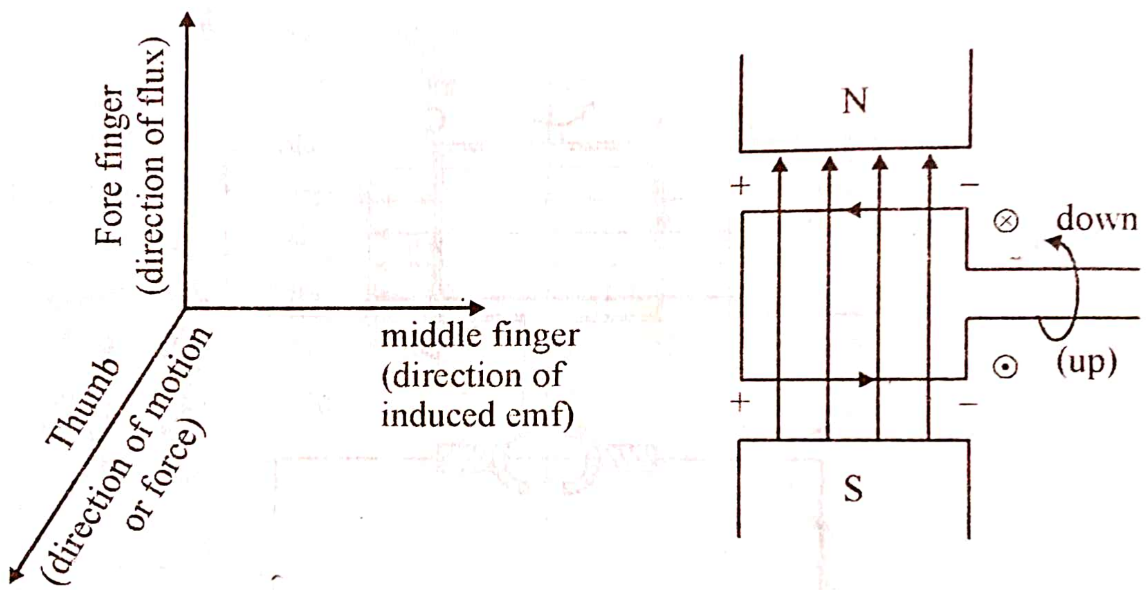

Fig. 5.5 Fleming's left Hand Rule & its application

The alternative expression for torque developed in a dc motor is as given under

$$\boxed{T = \frac{\phi I_a}{2\pi}} \quad \text{.....(5.6)}$$

Where, ϕ is the average flux per pole (Wb)

I_a is the current flowing in the coil (A)

Thus, torque developed in a dc motor is proportional to the product of flux and current.

The construction of practical dc machine is quite different from its single loop representation. In the following section the construction of a practical dc machine is briefly discussed.